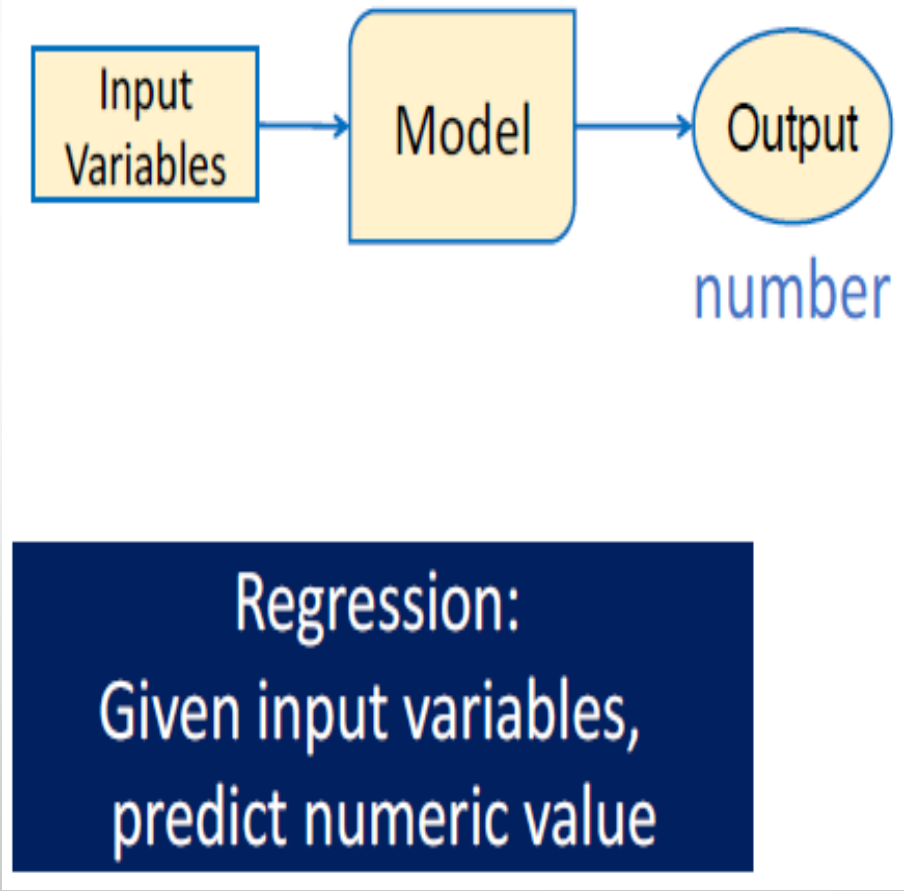
Regression , Linear Regression

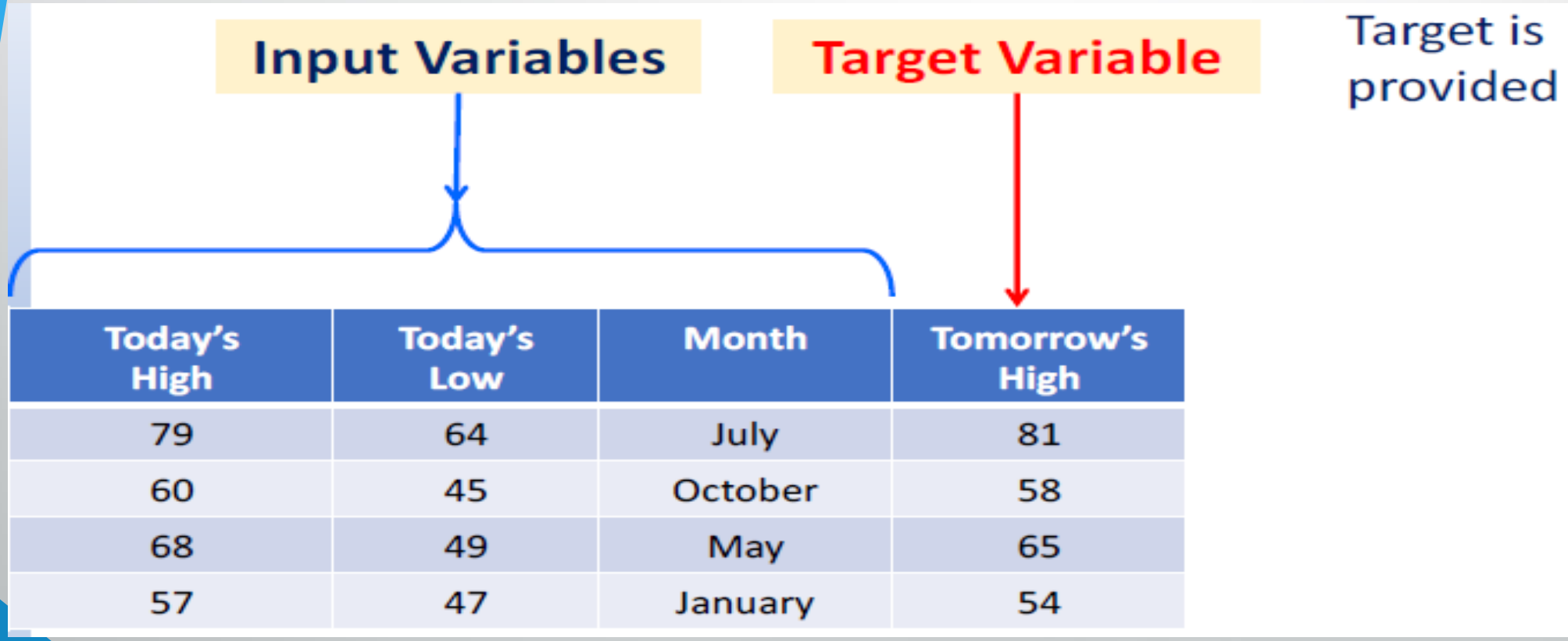
By Iqra Munwer

Regression

- When the model has to predict a numeric value then the task becomes a regression problem
- An example of regression is to predict the price of stock
- If you have to predict whether stock price will go up or down instead of numeric value it would be a classification task
- Examples:
  - Forecast high temperature for next day
  - Estimate average house price for a region
  - Determine demand for a new product
  - Predict power usage



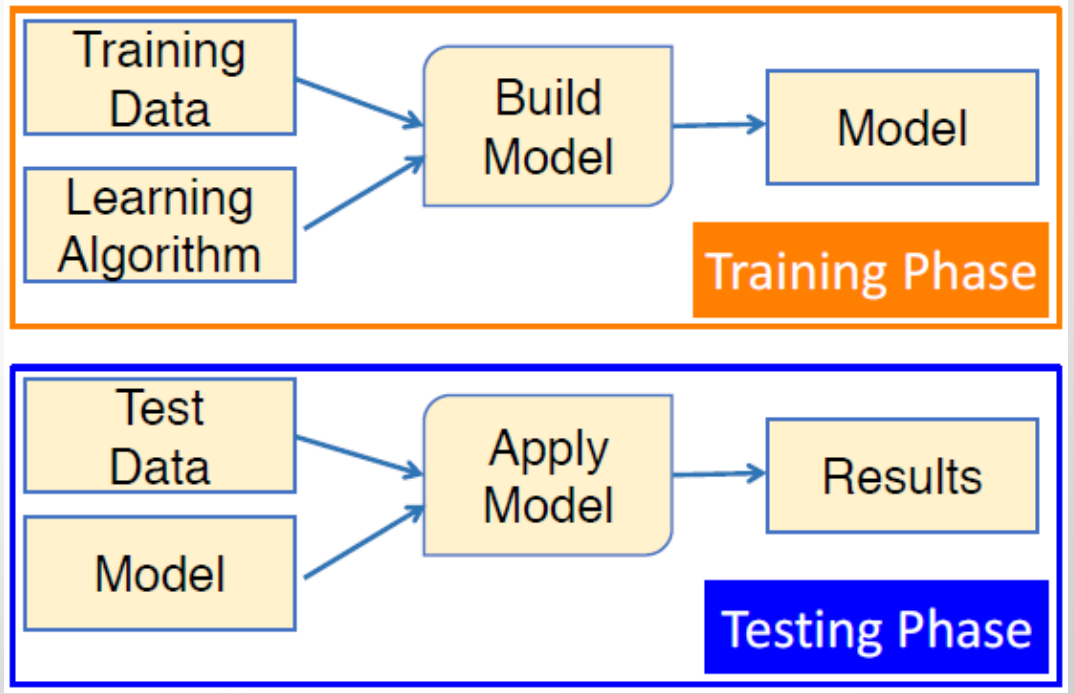
Regression is Supervised



Today's High	Today's Low	Month	Tomorrow's High
79	64	July	81
60	45	October	58
68	49	May	65
57	47	January	54

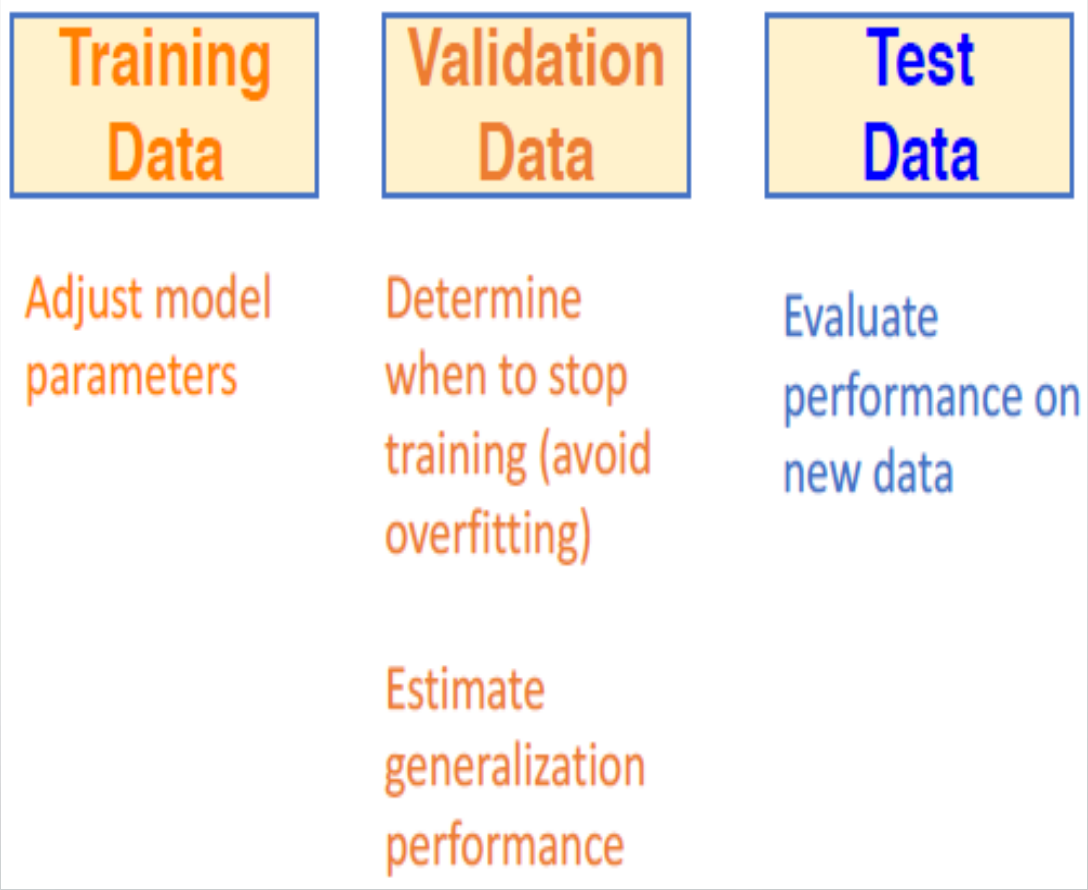
Training vs. Testing Phases

- As with classification, building a regression model also involves two phases
- A training phase in which the model is built
- A testing phase in which model is applied to new data that the model has not seen before



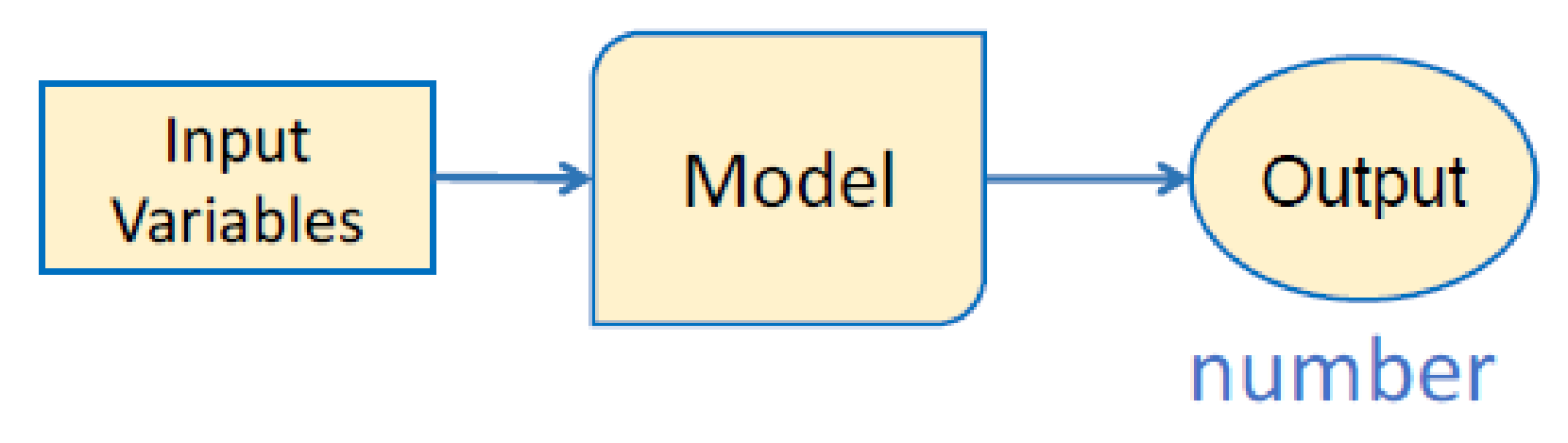
Datasets

- The training dataset is used to train the model i.e. to adjust the parameters of the model to learn the input to output mapping
- The test dataset is used to evaluate the performance of the model on new data
- The validation dataset is used to determine when stopping training in order to avoid model overfitting



Summary

- Predict number from input variables
- Regression is supervised task
- Target variable is numerical



Linear Regression

- Our objective is to study the relationship between two variables X and Y.
- One way is by means of **regression.**
- Regression analysis is the process of estimating a functional relationship between X and Y. A regression equation is often used to predict a value of Y for a given value of X.
- Another way to study relationship between two variables is **correlation.** It involves measuring the direction and the strength of the **linear** relationship.
- **Linear Regression** is a technique to model the relationship between a dependent variable (**Y**, like Pizza Price) and an independent variable (**X**, like Pizza Size) by fitting a linear equation to observed data.

Simplest Possible Linear Regression Model

- This is the base model for all statistical machine learning
- x is a one feature data variable
- y is the value we are trying to predict
- The general formula for simple linear regression is:
- $y = mx + cy \ = \ mx \ + c$
- Two parameters to estimate – the slope of the line m and the y-intercept c

Example Dataset: Pizza Size vs Price

Pizza Size (inches) (X)	Price (USD) (Y)
6	7
8	9
10	13
14	17.5
18	18

Step 1: Find the Mean (Average)

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{y} = \frac{\sum y}{n}$$

$$Sum\ of\ x\text{: } 6 + 8 + 10 + 14 + 18 = 566 + 8 + 10 + 14 + 18 = 56$$

$$Sum\ of\ y\text{: } 7 + 9 + 13 + 17.5 + 18 = 64.57 + 9 + 13 + 17.5 + 18 = 64.5$$

$$Number\ of\ data\ points\ n = 5$$

$$\bar{x} = \frac{56}{5} = 11.2$$

$$\bar{y} = \frac{64.5}{5} = 12.9$$

Step 2 Find the Slope (m)

x_i	y_i	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - x)^2$
6	7	-5.2	-5.9	30.68	27.04
8	9	-3.2	-3.9	12.48	10.24
10	13	-1.2	0.1	-0.12	1.44
14	17.5	2.8	4.6	12.88	7.84
18	18	6.8	5.1	34.68	46.24

Continue…

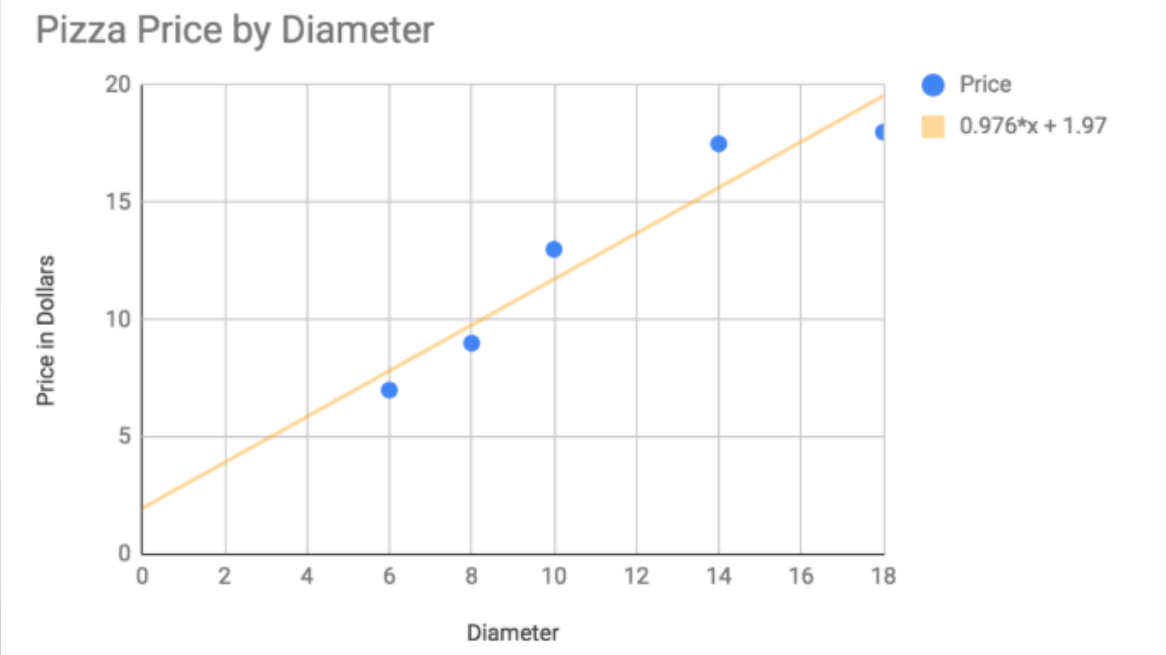
- $Now\ sum\ the\ columns$:
- $Sum\ of\ (xi - \bar{x})(yi - \bar{y}) = 30.68 + 12.48 - 0.12 + 12.88 + 34.68 = 90.6$
- $Sum\ of\ (xi - \bar{x})2 = 27.04 + 10.24 + 1.44 + 7.84 + 46.24 = 92.8$
- $Now,$
- $m = \frac{90.6}{92.8} = 0.976$

- Step 3: Find the Y-Intercept (c)
  - Formula: $c = \bar{y} - m\bar{x}$
  - Substituting Values : $c = 12.9 - (0.976)(11.2)$
  - $c = 1.9688$

Final Linear Regression Equation

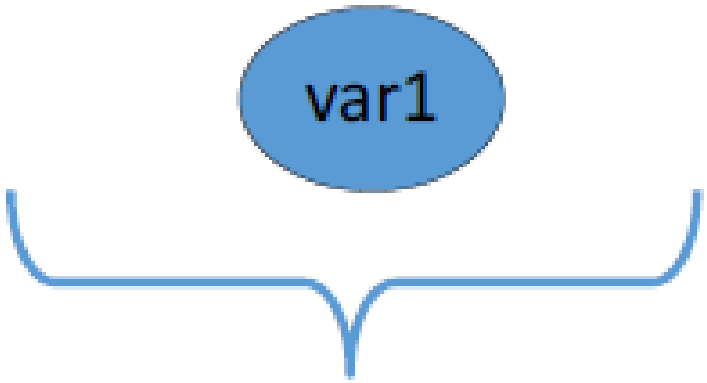
- Thus, the regression line equation is:
- $y = 0.976x + 1.97$
- **Making Predictions**
- Predict the price of a 12-inch pizza:
  - $Substituting\ x = 12$:
  - $y = 0.976(12) + 1.97$
  - $y = 11.712 + 1.97$
  - $y = 13.682$
- Thus, the predicted price of a **12-inch pizza** is **$13.68**.

Linear Regression Graph



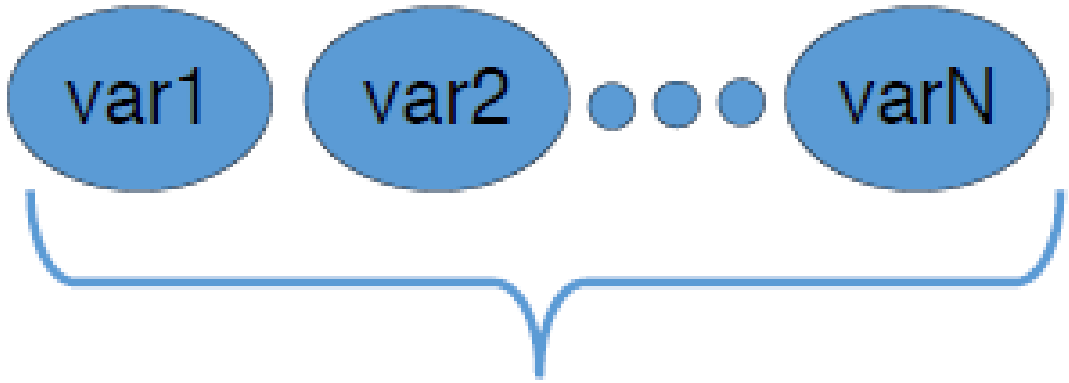
Types of Linear Regression

Simple Linear Regression



Input has one variable

Multiple Linear Regression



Input has >1 variables

Linear Regression Summary

- The linear regression captures the linear relationships between a numerical output and the input variables
- The least squares method can be used to build a linear regression model by finding the best-fitting line through the samples
- Discuss "European Soccer Regression Analysis using scikit-learn.ipynb"